

Data Communications and Networking Fourth Edition

Forouzan

Chapter 1

Introduction

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1-1 DATA COMMUNICATIONS

The term telecommunication means communication at a distance.

The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data.

Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.

- **Effective Data Communication depends on three fundamental characteristics**
  - **Delivery**
  - **Accuracy**
  - **Timeliness**

Topics discussed in this section:

- **Components of a data communications system**
- **Data Flow**

Figure 1.1 *Components of a data communication system*

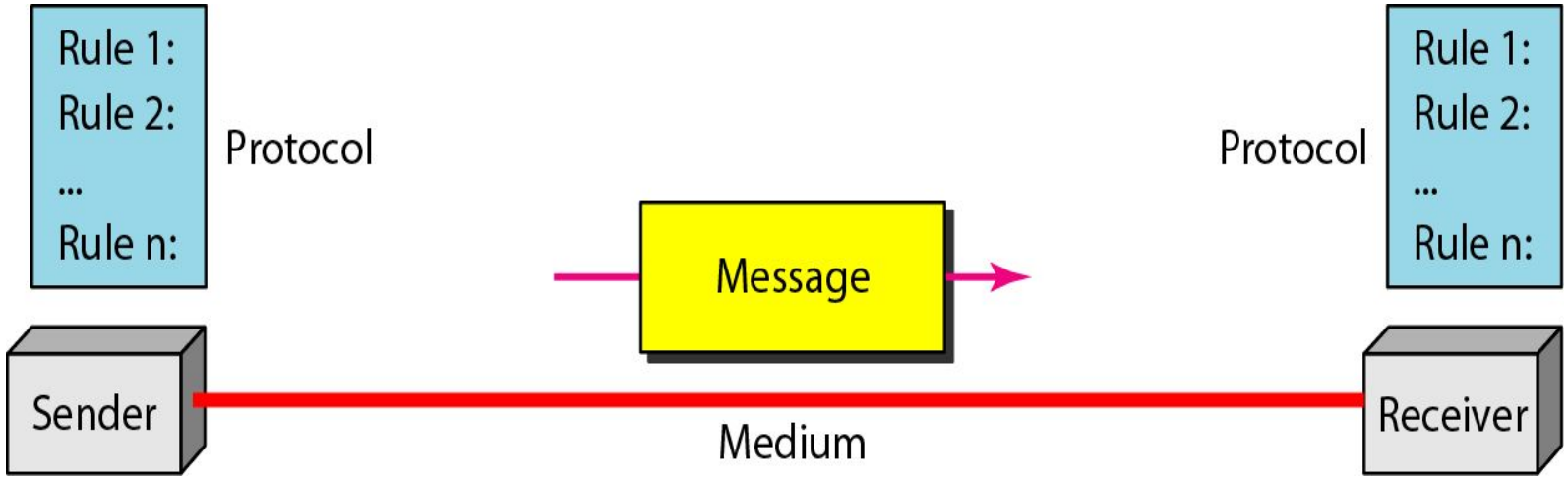
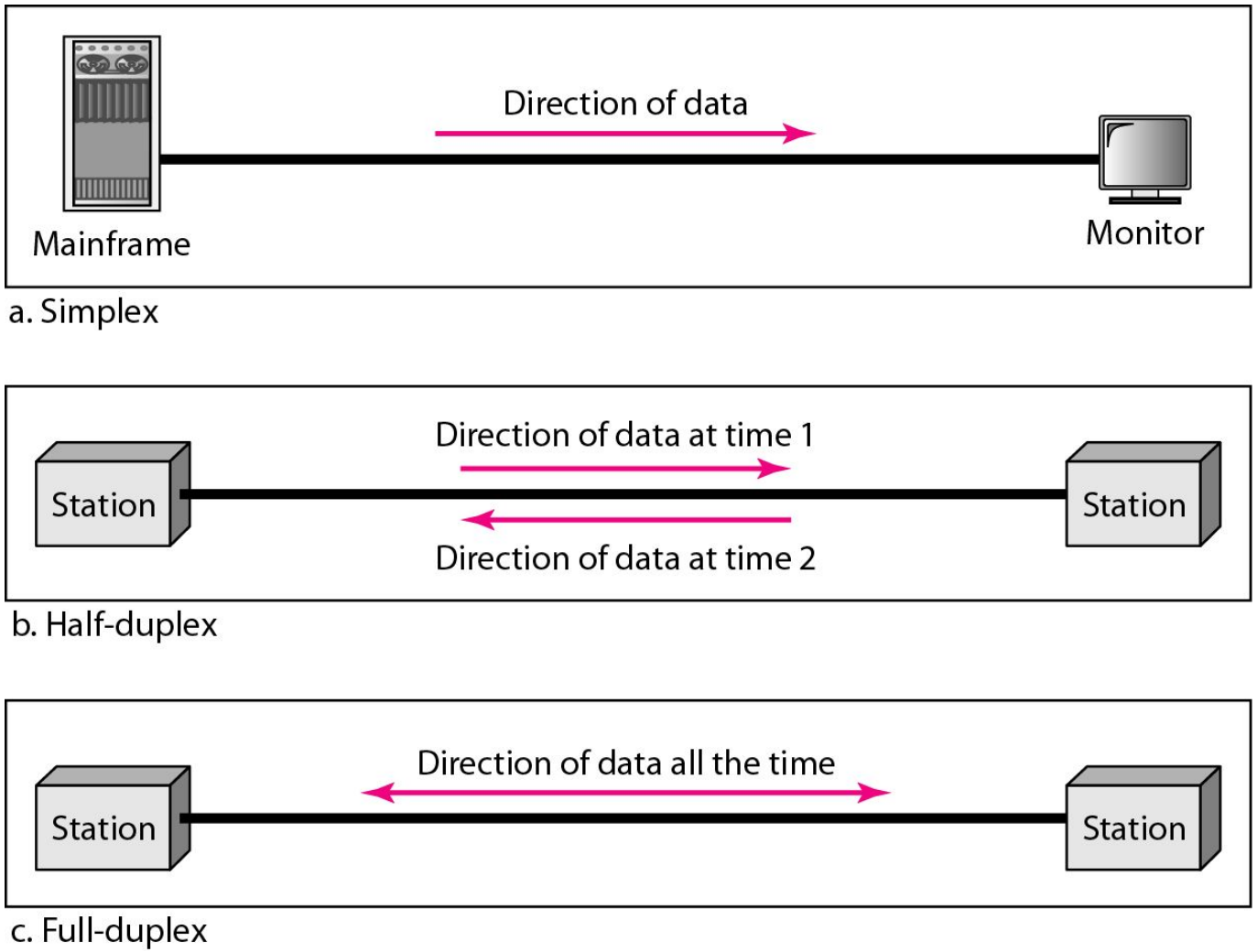


Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



1-2 NETWORKS

A network is a set of devices (often referred to as nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network. A link can be a cable, air, optical fiber, or any medium which can transport a signal carrying information.

Topics discussed in this section:

- **Network Criteria**
- **Physical Structures**
- **Categories of Networks**

Network Criteria

A network must meet following 3 criteria's

Performance

- The network-performance depends on following factors:
  i) Number of users
  ii) Type of transmission-medium
  iii) Efficiency of software

Reliability

- Failure rate of network components
- Measured in terms of availability/robustness
- More the failures are, less is the network's reliability.

Security

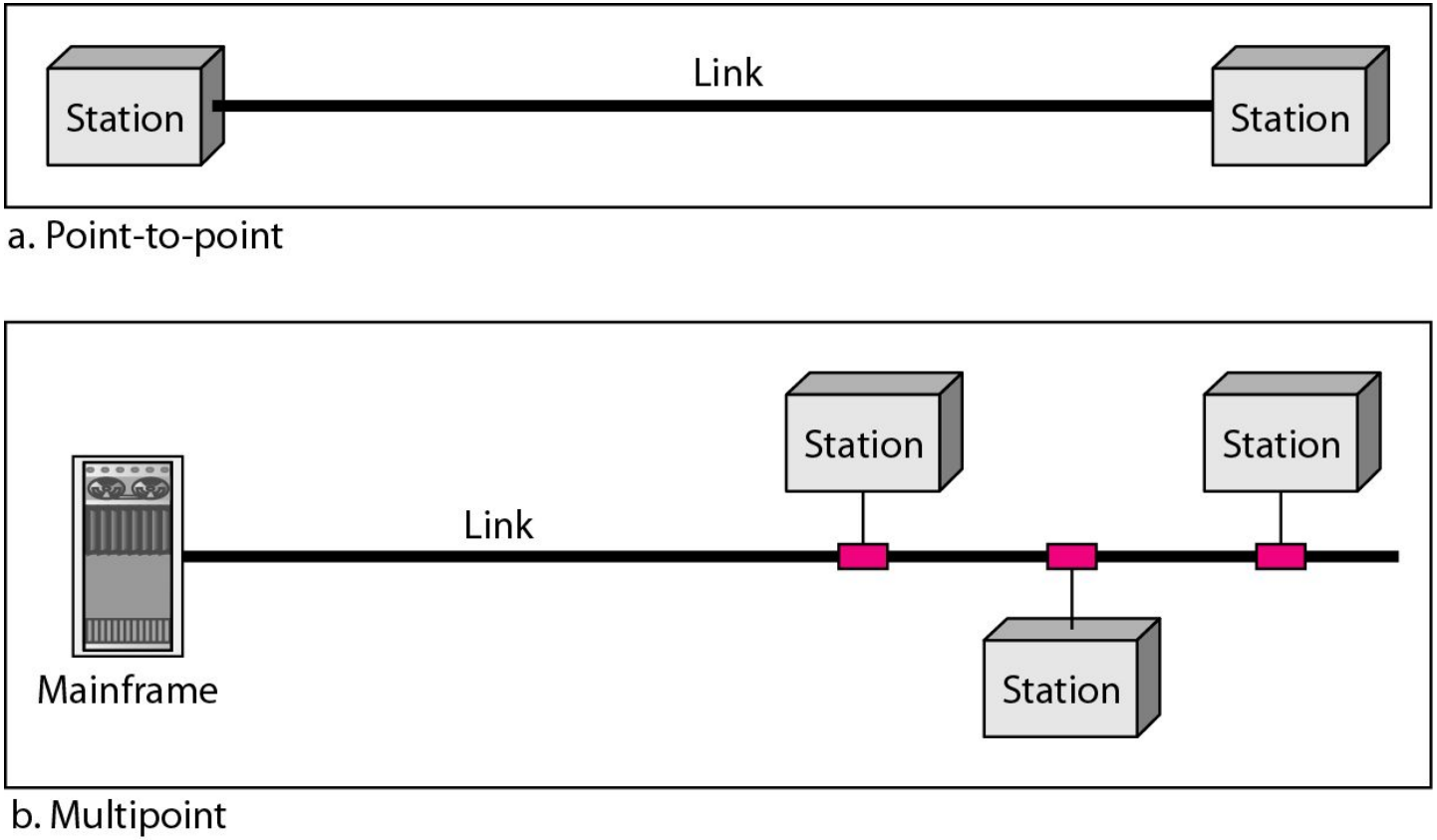
- **Data protection against corruption/loss of data due to:**
  - Errors
  - Malicious users

Physical Structures

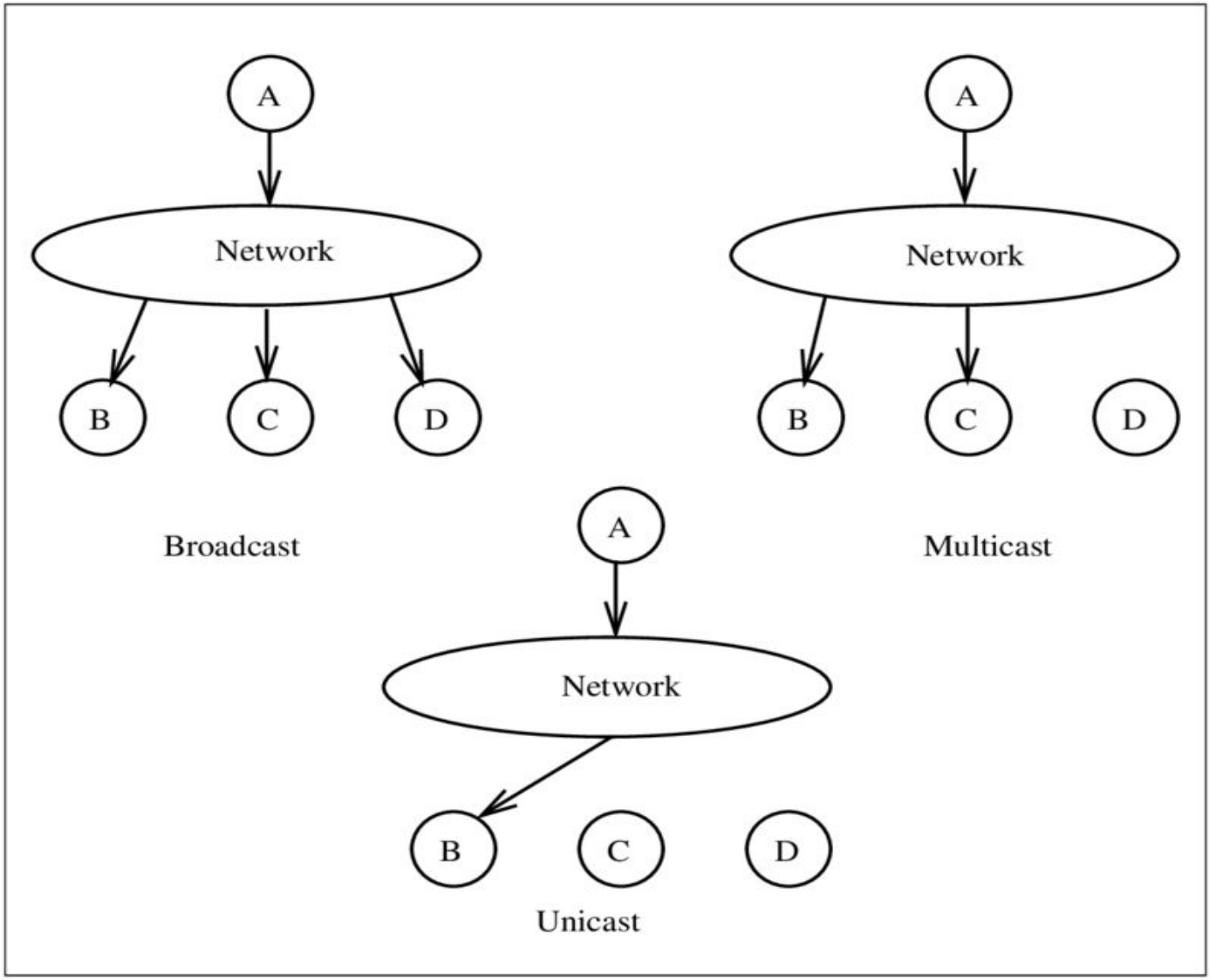
- **Type of Connection**
  - Point to Point - single transmitter and receiver
  - Multipoint - multiple recipients of single transmission

- **Physical Topology**
  - Connection of devices
  - Type of transmission - unicast, mulitcast, broadcast

Figure 1.3 *Types of connections: point-to-point and multipoint*



Type of transmission - unicast, multicast, broadcast



Network Devices

- **To communicate data through different transmission media and to configure networks with different functionality, we require different devices like**
  - Hub
  - Bridge
  - Switch
  - Router

https://www.youtube.com/watch?v=eMamgWllRFY&t=29s

Figure 1.4 *Categories of topology*

The physical-topology defines how devices are connected to make a network.

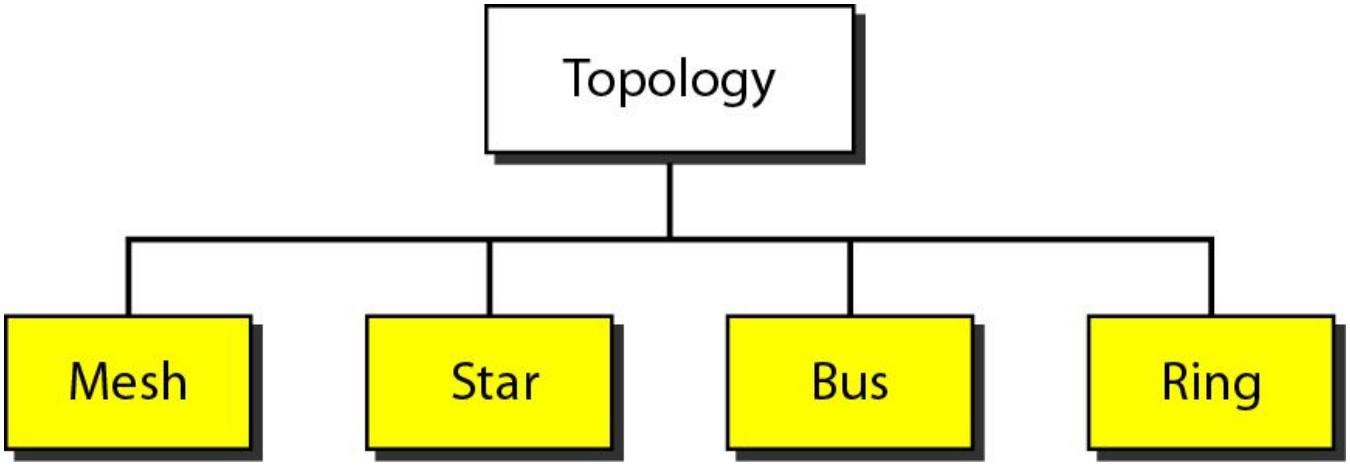
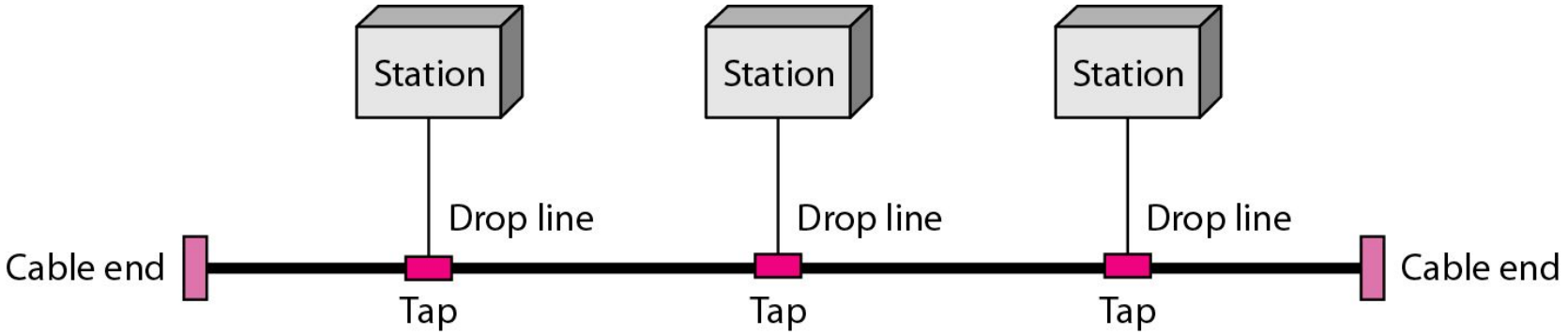


Figure 1.7 *A bus topology connecting three stations*

- All the devices are connected to the single cable called bus.
- Every device communicates with the other device through this bus.
- A data from the source is broadcasted to all devices connected to the bus.
- Only the intended-receiver, whose physical-address matches, accepts the data.
- Devices are connected to the bus by drop-lines and taps.
- A drop-line is a connection running between the device and the bus.
- A tap is a connector that links to the bus



•Advantages

1) Easy installation.

2) Cable required is the least compared to mesh/star topologies.

3) Redundancy is eliminated.

4) Costs less (Compared to mesh/star topologies). 5) Mostly used in small networks. Good for LAN.

• Disadvantages

1) Difficult to detect and troubleshoot fault.

2) Signal reflection at the taps can cause degradation in quality.

3) A fault/break in the cable stops all transmission.

4) There is a limit on i) Cable length ii) Number of nodes that can be connected.

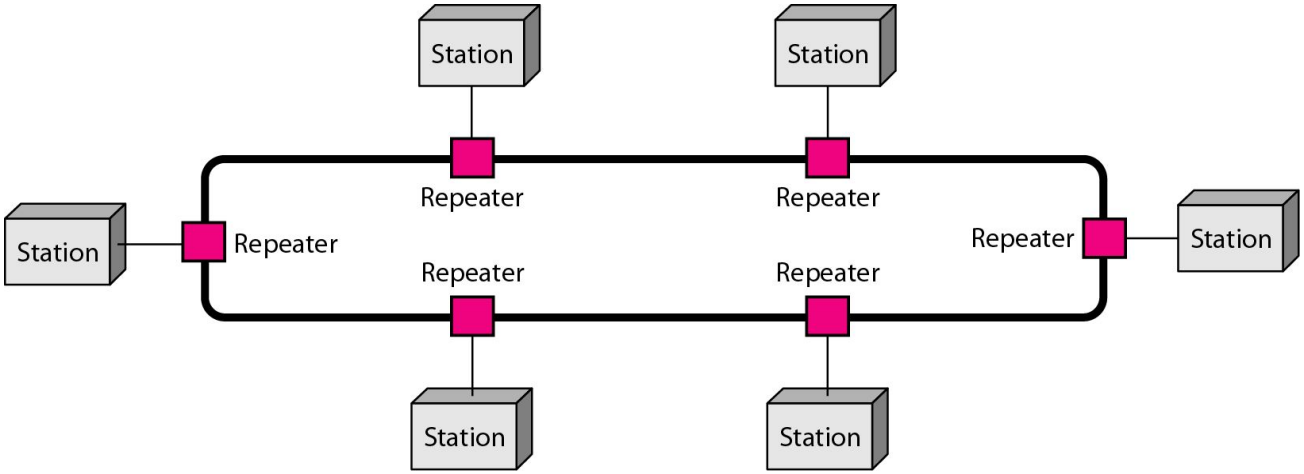
5) Security is very low because all the devices receive the data sent from the source.

Figure 1.8 *A ring topology connecting six stations*

- Each device is connected to the next, forming a ring
- There are only two neighbors for each device.
- Data travels around the network in one direction till the destination is reached.
- Sending and receiving of data takes place by the help of token.
- Each device has a repeater.
- A repeater

→ receives a signal on transmission-medium &

→ regenerates & passes the signal to next device.



•Advantages:

1) Easy installation and reconfiguration.

To add/delete a device, requires changing only 2 connections.

3) Fault isolation is simplified.

If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network-operator to the problem and its location.

3)Congestion reduced: Because all the traffic flows in only one direction.

• Disadvantages:

1) Unidirectional traffic.

2) A fault in the ring/device stops all transmission.

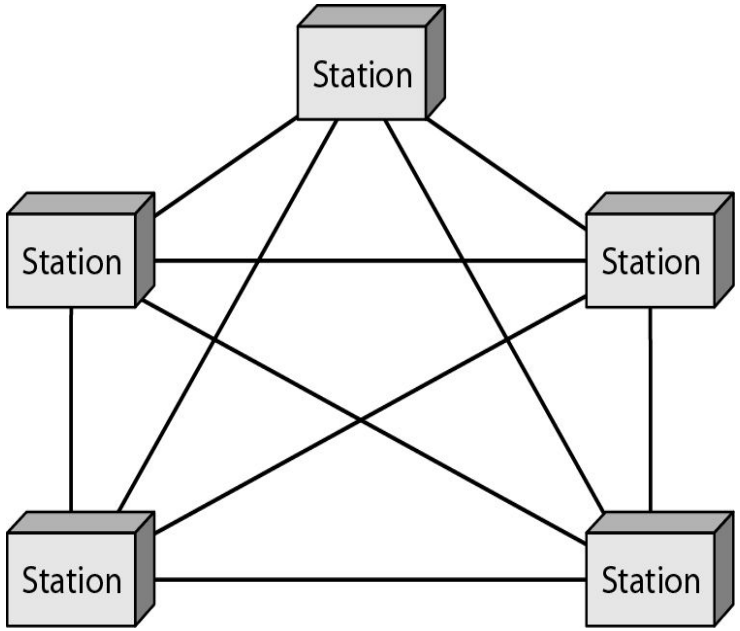
3) There is a limit on

i) Cable length & ii) Number of nodes that can be connected.

4) Slower: Each data must pass through all the devices between source and destination.

Figure 1.5 *A fully connected mesh topology (five devices)*

- All the devices are connected to each other
- There exists a dedicated point-to-point link between all devices.
- There are n(n-1) physical channels to link n devices.
- Every device not only sends its own data but also relays data from other nodes.
- For 'n' nodes,
  → there are n(n-1) physical-links



- **Advantages:**

1) Congestion reduced: Each connection can carry its own data load.

2) Robustness: If one link fails, it does not affect the entire system.

3) Security: When a data travels on a dedicated-line, only intended-receiver can see the data.

4) Easy fault identification & fault isolation: Traffic can be re-routed to avoid problematic links.

- **Disadvantages:**

1) Difficult installation and reconfiguration.

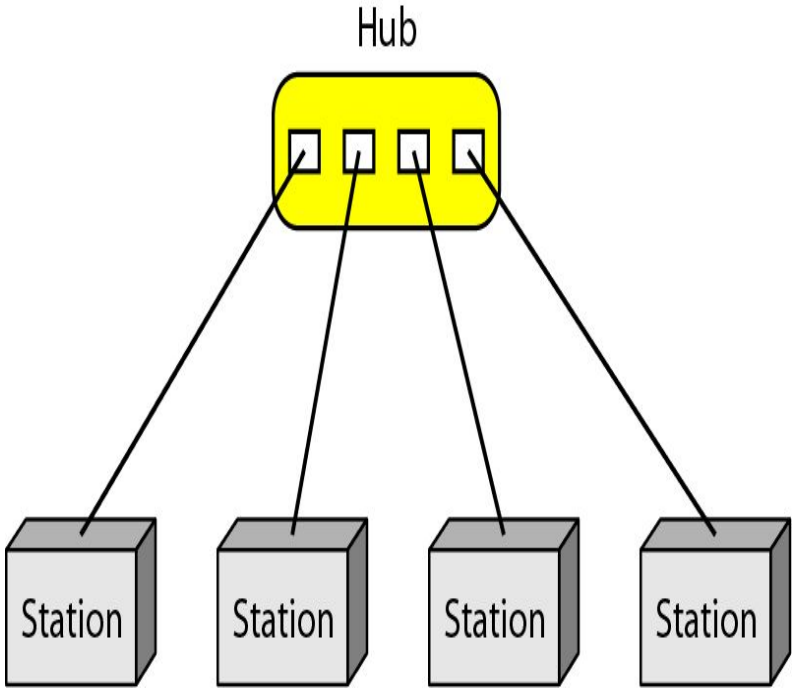
2) Bulk of wiring occupies more space than available space.

3) Very expensive: as there are many redundant connections.

4) Not mostly used in computer networks

Figure 1.6 *A star topology connecting four stations*

- All the devices are connected to a central controller called a hub.
- There exists a dedicated point-to-point link between a device & a hub.
- The devices are not directly linked to one another. Thus, there is no direct traffic between devices.
- The hub acts as a junction: If device-1 wants to send data to device-2, the device-1 sends the data to the hub, then the hub relays the data to the device-2.



•Advantages:

1) Less expensive: Each device needs only one link & one I/O port to connect it to any devices.

2) Easy installation & reconfiguration: Nodes can be added/removed which are affecting the network.

3) Robustness: If one link fails, it does not affect the entire system.

4) Easy to detect and troubleshoot fault.

5) Centralized management: The hub manages and controls the whole network.

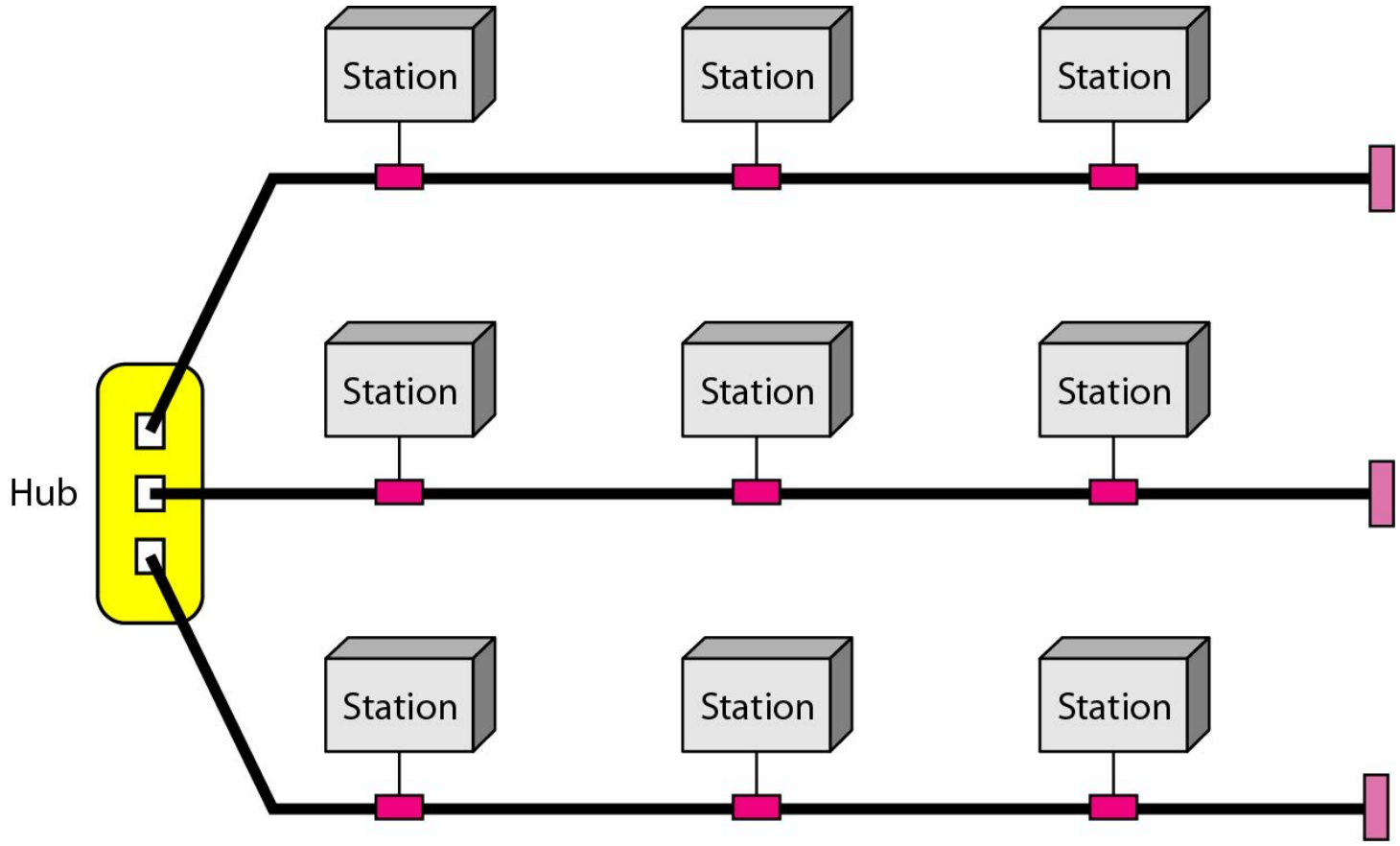
• Disadvantages:

1) Single point of failure: If the hub goes down, the whole network is dead.

2) Cable length required is the more compared to bus/ring topologies.

3) Number of nodes in network depends on capacity of hub.

Figure 1.9 *A hybrid topology: a star backbone with three bus networks*



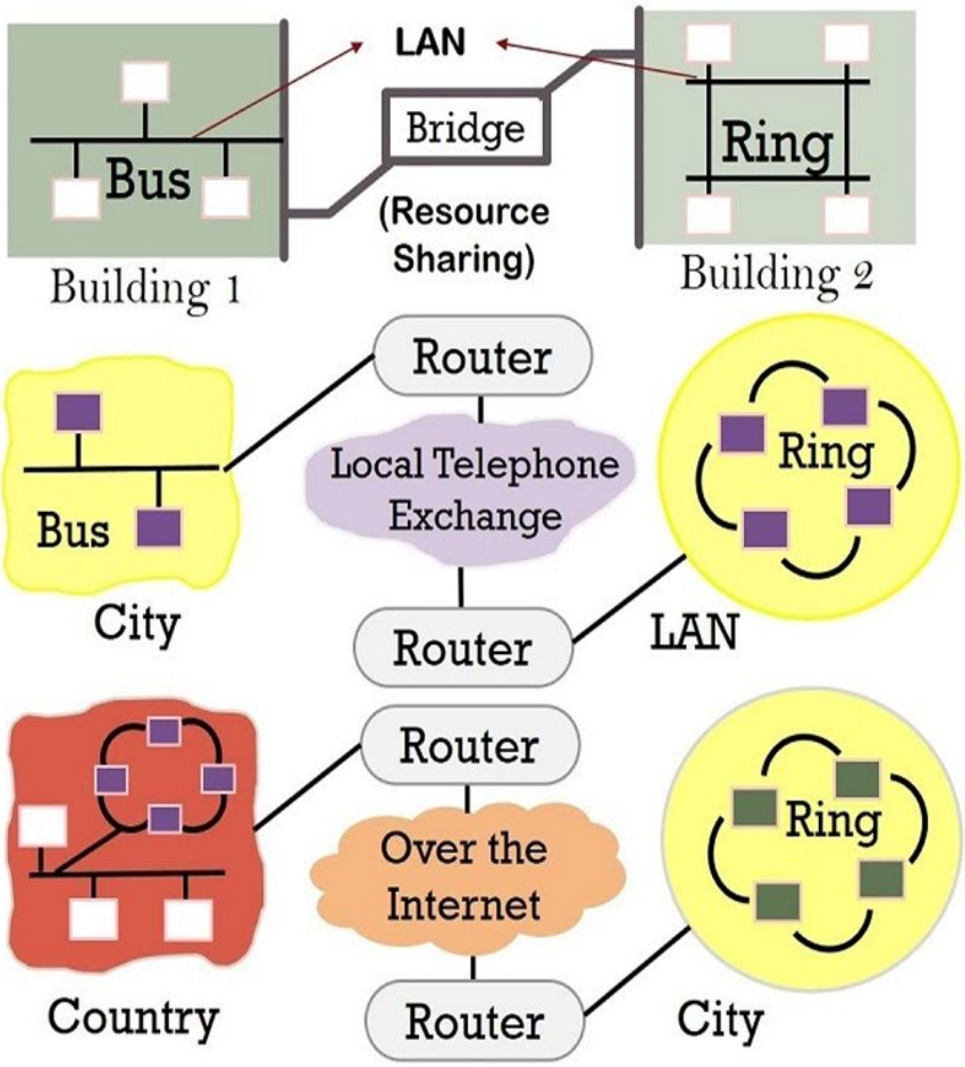
Categories of Networks

- **Local Area Networks (LANs)**
  - Short distances
  - Designed to provide local interconnectivity
- **Wide Area Networks (WANs)**
  - Long distances
  - Provide connectivity over large areas
- **Metropolitan Area Networks (MANs)**
  - Provide connectivity over areas such as a city, a campus

LAN

MAN

WAN


LAN
Bus
Bridge
(Resource Sharing)
Ring
Building 1
Building 2
Router
Bus
Local Telephone Exchange
Ring
City
Router
LAN
Router
Over the Internet
Ring
Country
Router
City

Parameters	LAN	WAN
Expands to	Local Area Network	Wide Area Network
Meaning	LAN is used to connect computers in a single office, building or campus	WAN is used to connect computers in a large geographical area such as countries
Ownership of network	Private	Private or public
Range	Small: up to 10 km	Large: Beyond 100 km
Speed	High: Typically 10, 100 and 1000 Mbps	Low: Typically 1.5 Mbps
Propagation Delay	Short	Long
Cost	Low	High
Congestion	Less	More
Design & maintenance	Easy	Difficult
Fault Tolerance	More Tolerant	Less Tolerant
Media used	Twisted pair	Optical fiber or radio waves
Used for	College, Hospital	Internet
Interconnects	LAN interconnects hosts	WAN interconnects connecting devices such as switches, routers, or modems

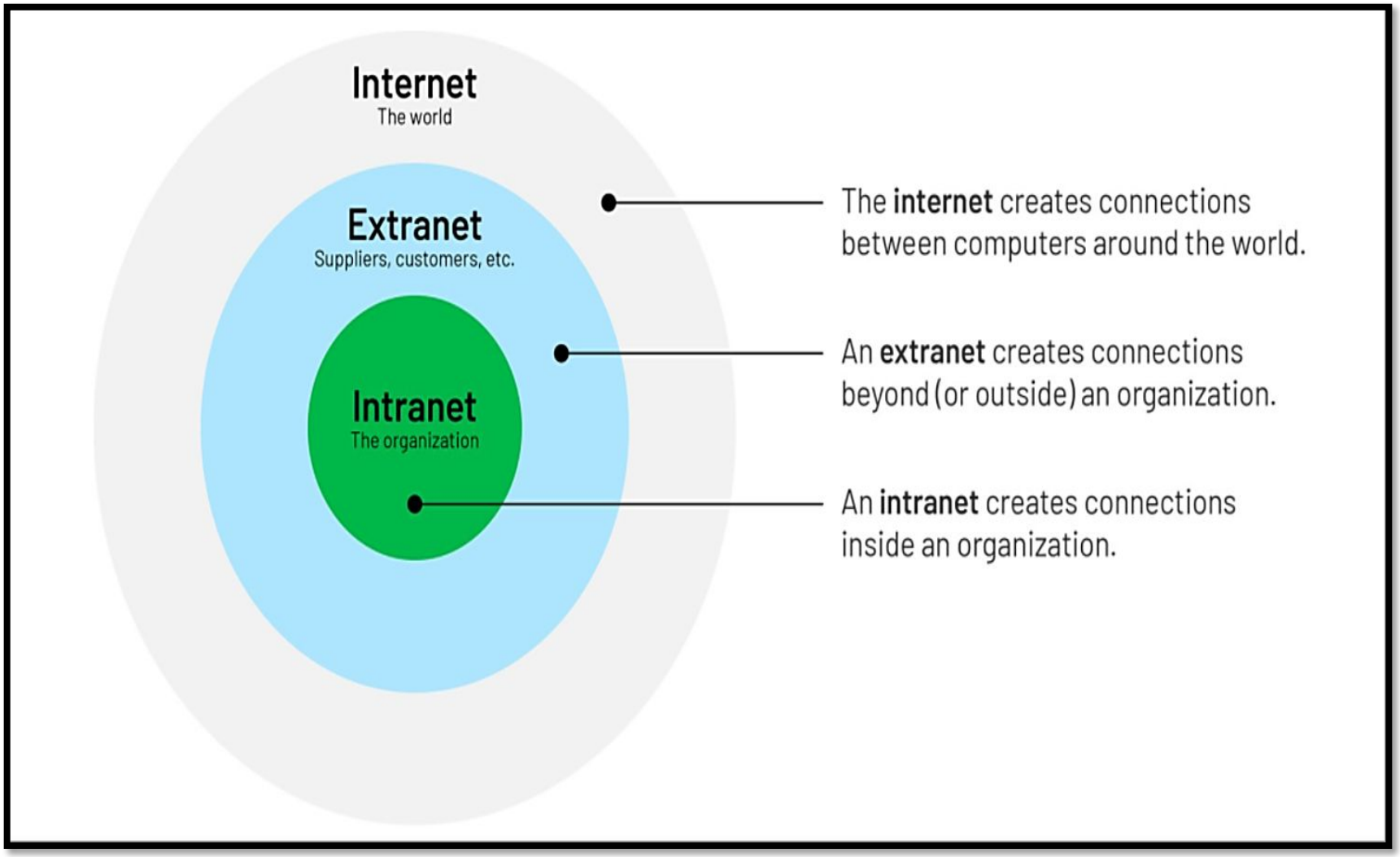
1-3 THE INTERNET

The Internet is **a vast network that connects computers all over the world**. Through the Internet, people can share information and communicate from anywhere with an Internet connection.

A network of networks is called an internet

Topics discussed in this section:

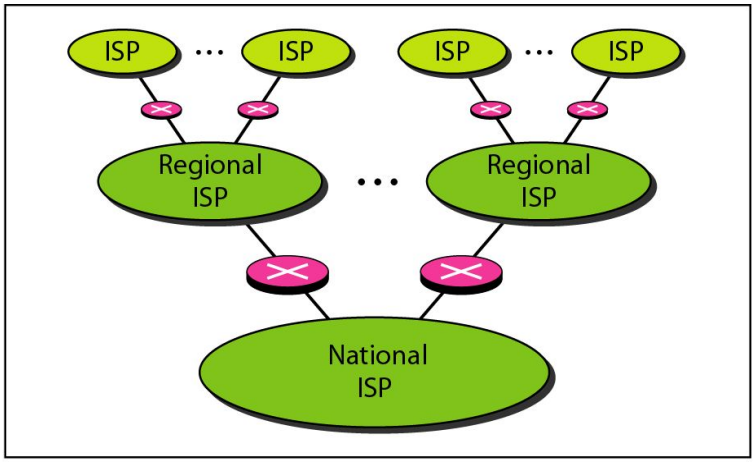
Organization of the Internet
Internet Service Providers (ISPs)


Internet
The world
Extranet
Suppliers, customers, etc.
Intranet
The organization
The internet creates connections between computers around the world.
An extranet creates connections beyond (or outside) an organization.
An intranet creates connections inside an organization.

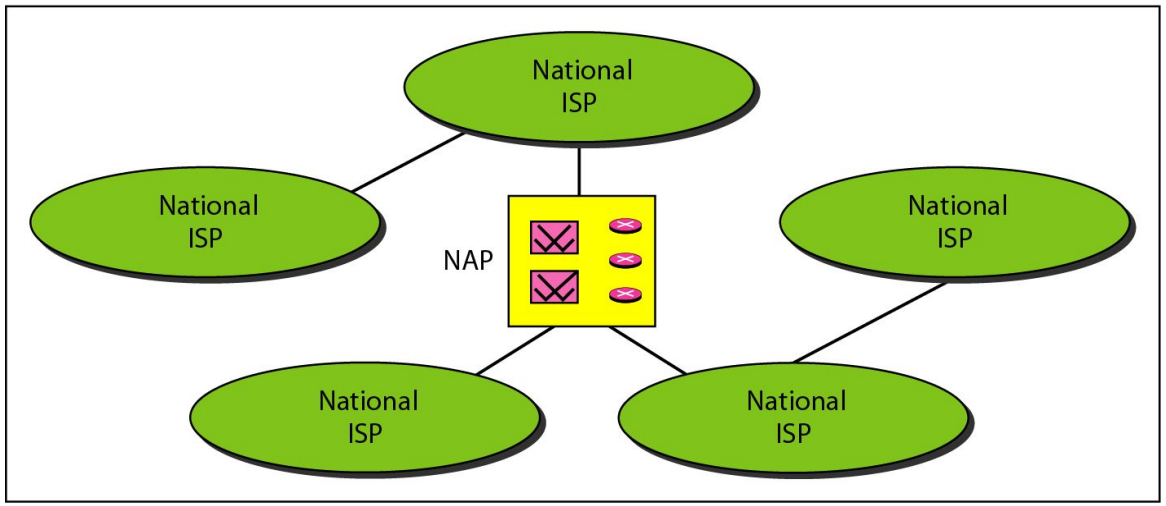
ISP stands for Internet Service Provider

A Regional ISP provides internet access to a specific area and usually has a smaller technical support team than National ISP which is a business that provides internet access in cities and towns nationwide and have a much larger technical support team.

Figure 1.13 *Hierarchical organization of the Internet*



a. Structure of a national ISP



b. Interconnection of national ISPs

When you type in a web address into your browser...

Step 1: Your PC or device is connected to the web through a modem or router. Together, these devices allow you to connect to other networks around the globe.
Your router enables multiple computers to join the same network while a modem connects to your ISP (Internet Service Provider) which provides you with either cable or DSL (Digital Subscriber Line) internet.

Step 2: Type in a web address, known as a URL (Uniform Resource Locator). Each website has its own unique URL that signals to your ISP where you want to go.

Step 3: Your query is pushed to your ISP which connects to several servers which store and send data like.
Next, your browser looks up the IP address for the domain name you typed into your search engine through DNS. DNS then translates the text-based domain name you type into the browser into the number-based IP address.

- Example: Google.com becomes 64.233.191.255

Step 4: Your browser sends a Hypertext Transfer Protocol (HTTP) request to the target server to send a copy of the website to the client using TCP/IP.

Step 5: The server then approves request and sends a "200 OK" message to your computer. Then, the server sends website files to the browser in the form of data packets.

Step 6: As your browser reassembles the data packets, the website loads allowing you to learn, shop, browse, and engage.

Step 7: Enjoy your search results!

https://developer.mozilla.org/en-US/docs/Web/HTTP/Status#successful_responses

https://www.youtube.com/watch?v=x3c1ih2NJEg

Internet Standards

- An **Internet standard** is a thoroughly tested specification that is useful to and adhered to by those who work with the Internet.
- It is a formalized regulation that must be followed.
- There is a strict procedure by which a specification attains Internet standard status. A specificationbegins as an Internet draft. An **Internet draft** is a working document (a work in progress) with no official status and a six-month lifetime.
- Upon recommendation from the Internet authorities, a draft may be published as a **Request for Comment (RFC).**
- Each RFC is edited, assigned a number, and made available to all interested parties. RFCs go through maturity levels and are categorized according to their requirement level.

Internet Standards

Maturity Levels

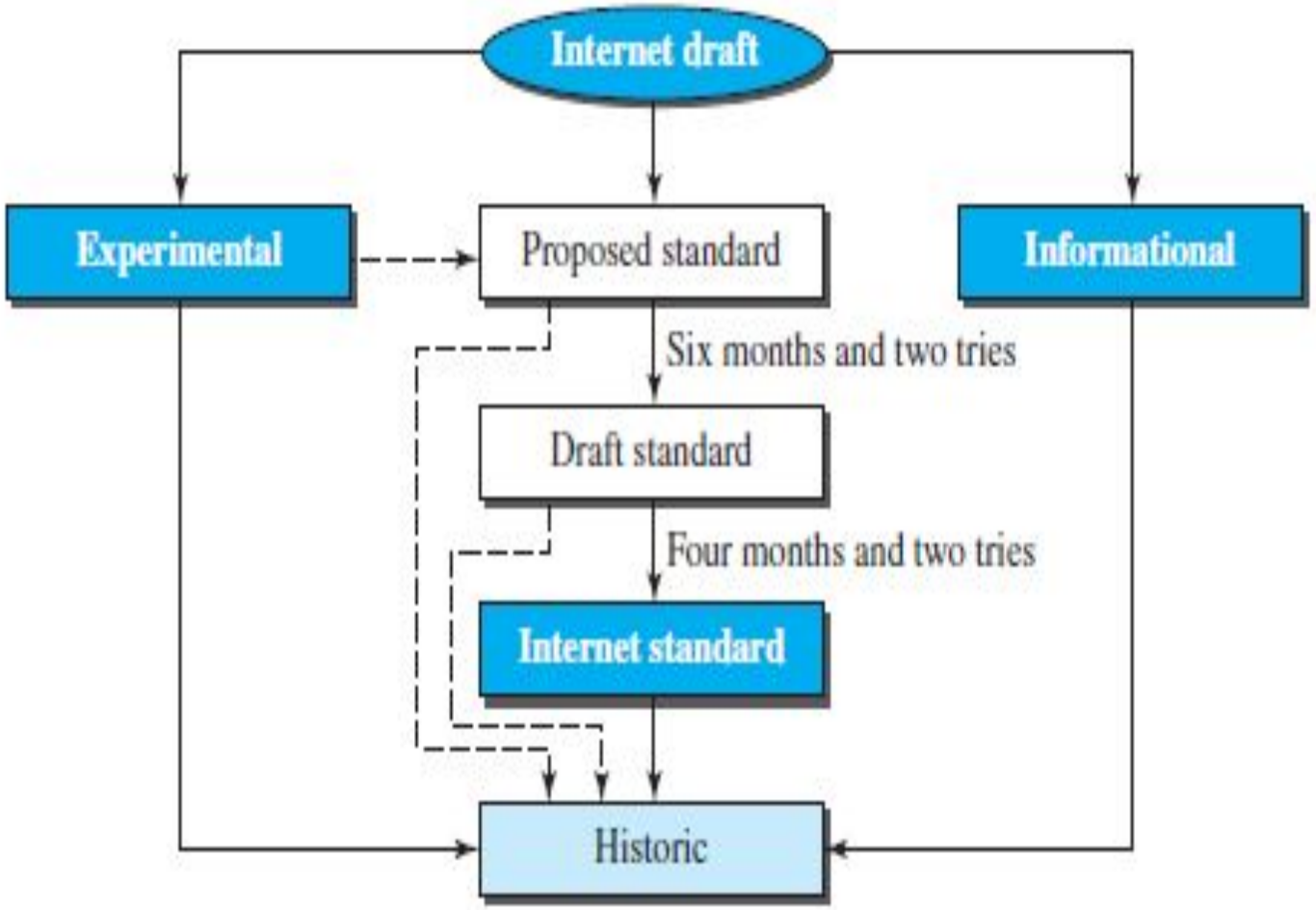
- An RFC, during its lifetime, falls into one of six *maturity levels:* proposed standard, draft
- standard, Internet standard, historic, experimental, and informational (see Figure 1.16).
- ***Proposed Standard*** **.** A proposed standard is a specification that is stable, well understood, and of sufficient interest to the Internet community. At this level, the specification is usually tested and implemented by several different groups.
- ❑ Draft Standard. A proposed standard is elevated to draft standard status after at least two successful independent and interoperable implementations.
- Barring difficulties, a draft standard, with modifications if specific problems are encountered, normally becomes an Internet standard.
- ❑ Internet Standard. A draft standard reaches Internet standard status after demonstrations of successful implementation.

Internet Standards

- ❑ Historic. The historic RFCs are significant from a historical perspective. They
- either have been superseded by later specifications or have never passed the necessary
- maturity levels to become an Internet standard.
- ❑ Experimental. An RFC classified as experimental describes work related to an
- experimental situation that does not affect the operation of the Internet. Such an
- RFC should not be implemented in any functional Internet service.
- ❑ Informational. An RFC classified as informational contains general, historical, or
- tutorial information related to the Internet. It is usually written by someone in a
- non-Internet organization, such as a vendor.

Internet Standards

Figure 1.16 *Maturity levels of an RFC*



- ***Requirement Levels***
- RFCs are classified into five *requirement levels:* required, recommended, elective, limited use, and not recommended.
- ❑ ***Required.*** An RFC is labeled *required* if it must be implemented by all Internet systems to achieve minimum conformance. For example, IP and ICMP are required protocols.
- ❑ ***Recommended.*** An RFC labeled recommended is not required for minimum conformance; it is recommended because of its usefulness.
- Example, FTP and TELNET are recommended protocols.
- ❑ ***Elective.*** An RFC labeled elective is not required and not recommended. However, a system can use it for its own benefit.

- ***Requirement Levels***

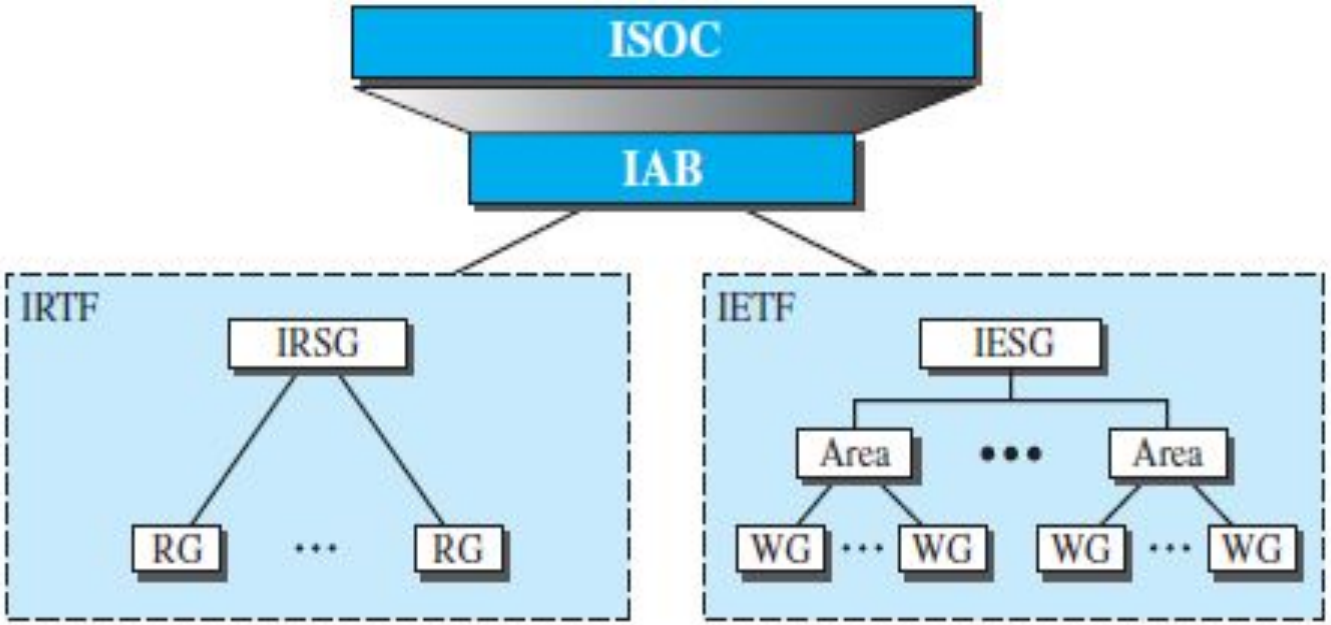
❑ **Limited Use. An RFC labeled limited use should be used only in limited situations. Most of the experimental RFCs fall under this category.**

❑ **Not Recommended. An RFC labeled not recommended is inappropriate for general use. Normally a historic (deprecated) RFC may fall under this category.**

Internet Administration

- The Internet, with its roots primarily in the research domain, has evolved and gained a broader user base with significant commercial activity.

Figure 1.17 *Internet administration*



- ***ISOC***
- The **Internet Society (ISOC)** is an international, nonprofit organization formed in
- 1992 to provide support for the Internet standards process. ISOC accomplishes this
- through maintaining and supporting other Internet administrative bodies such as IAB,
- IETF, IRTF, and IANA (see the following sections). ISOC also promotes research and
- other scholarly activities relating to the Internet.

Internet Administration

- ***IAB***
- The **Internet Architecture Board (IAB)** is the technical advisor to the ISOC. The
- main purposes of the IAB are to oversee the continuing development of the TCP/IP
- Protocol Suite and to serve in a technical advisory capacity to research members of the
- Internet community. IAB accomplishes this through its two primary components, the
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- Another responsibility of the IAB is the editorial management of the RFCs
- IAB is also the external liaison between the Internet and other standards organizations and forums

Internet Administration

- ***IETF***
- The **Internet Engineering Task Force (IETF)** is a forum of working groups managed by the Internet Engineering Steering Group (IESG).

- IETF is responsible for identifying operational problems and proposing solutions to these problems. IETF also develops and reviews specifications intended as Internet standards.

- The working groups are collected into areas, and each area concentrates on a specific topic. Currently nine areashave been defined.

- The areas include applications, protocols, routing, network management next generation (IPng), and security.

Internet Administration

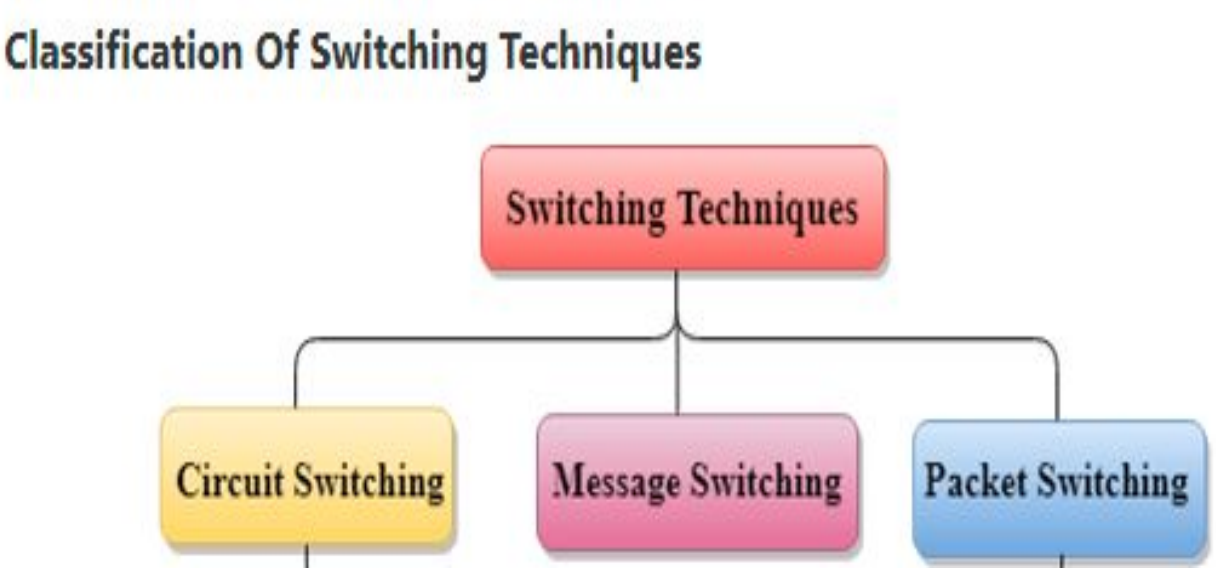
- ***IRTF***
- The **Internet Research Task Force (IRTF)** is a forum of working groups managed by the Internet Research Steering Group (IRSG).

- IRTF focuses on long-term research topic related to Internet protocols, applications, architecture, and technology.

Switching techniques

In large networks, there can be multiple paths from sender to receiver. The switching technique will decide the best route for data transmission.

Switching technique is used to connect the systems for making one-to-one communication.

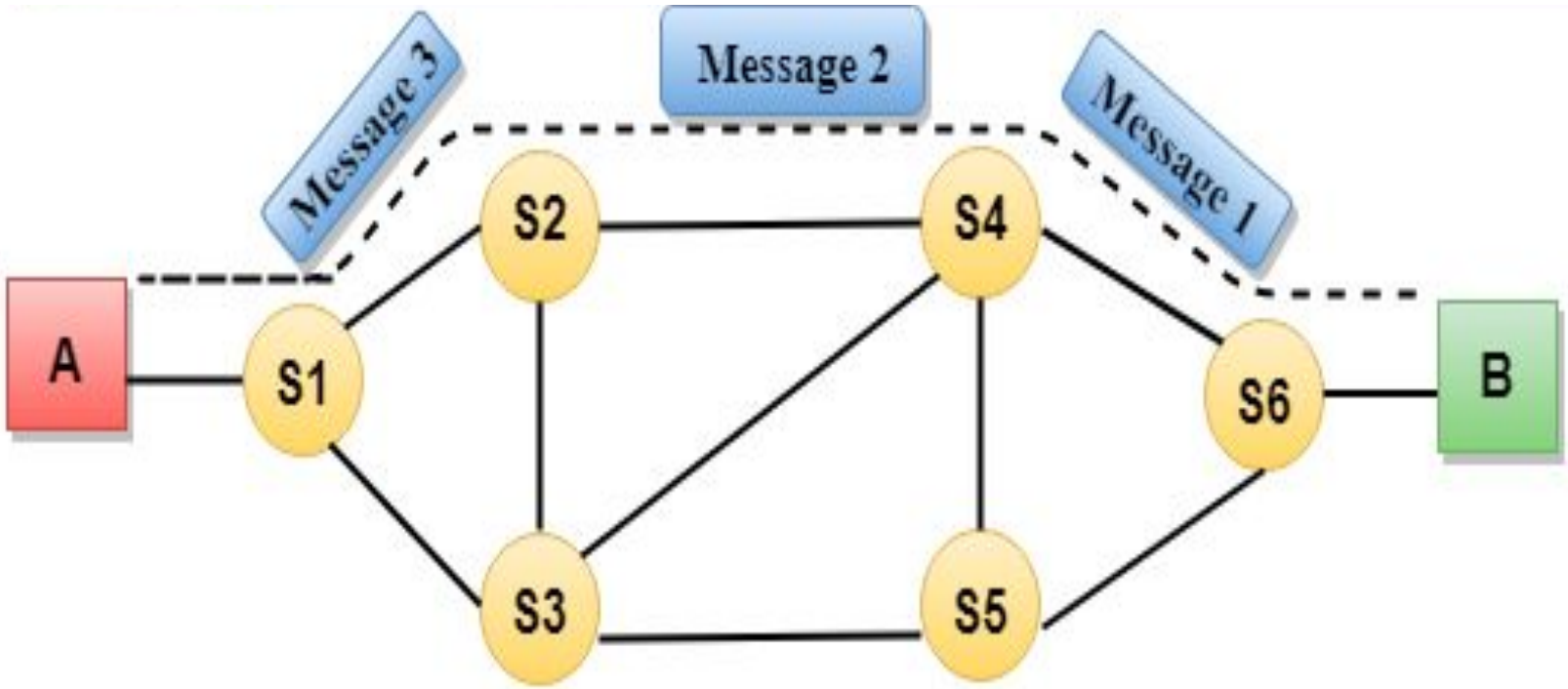


Circuit Switching

- Circuit switching is a switching technique that establishes a dedicated path between sender and receiver.
- In the Circuit Switching Technique, once the connection is established then the dedicated path will remain to exist until the connection is terminated.
- Circuit switching in a network operates in a similar way as the telephone works.
- A complete end-to-end path must exist before the communication takes place.
- In case of circuit switching technique, when any user wants to send the data, voice, video, a request signal is sent to the receiver then the receiver sends back the acknowledgment to ensure the availability of the dedicated path. After receiving the acknowledgment, dedicated path transfers the data.
- Circuit switching is used in public telephone network. It is used for voice transmission.
- Fixed data can be transferred at a time in circuit switching technology.

Communication through circuit switching has 3 phases:

- Circuit establishment
- Data transfer
- Circuit Disconnect


Message 3
Message 2
Message 1
A
S1
S2
S3
S4
S5
S6
B

Advantages Of Circuit Switching:

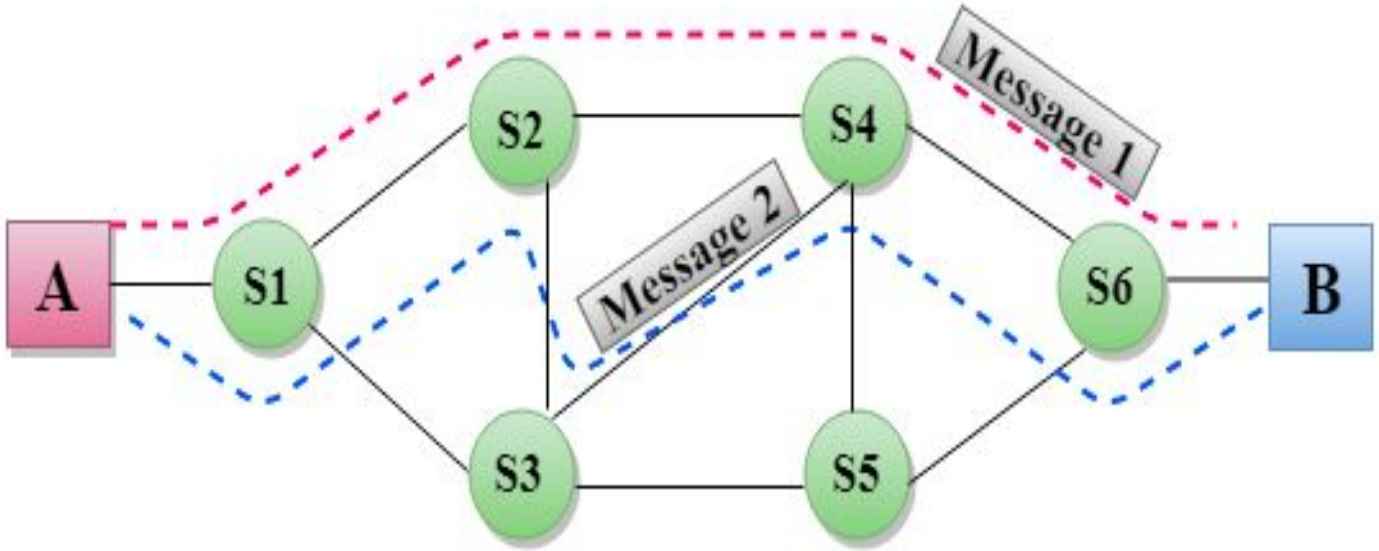
- In the case of Circuit Switching technique, the communication channel is dedicated.
- It has fixed bandwidth.

Disadvantages Of Circuit Switching:

- Once the dedicated path is established, the only delay occurs in the speed of data transmission.
- It takes a long time to establish a connection approx 10 seconds during which no data can be transmitted.
- It is more expensive than other switching techniques as a dedicated path is required for each connection.
- It is inefficient to use because once the path is established and no data is transferred, then the capacity of the path is wasted.
- In this case, the connection is dedicated therefore no other data can be transferred even if the channel is free.

Message Switching

- Message Switching is a switching technique in which a message is transferred as a complete unit and routed through intermediate nodes at which it is stored and forwarded.
- In Message Switching technique, there is no establishment of a dedicated path between the sender and receiver.
- The destination address is appended to the message. Message Switching provides a dynamic routing as the message is routed through the intermediate nodes based on the information available in the message.
- Message switches are programmed in such a way so that they can provide the most efficient routes.
- Each and every node stores the entire message and then forward it to the next node. This type of network is known as **store and forward network.**
- Message switching treats each message as an independent entity.


Message 1
S2
S4
Message 2
A
S1
S6
B
S3
S5

Advantages Of Message Switching

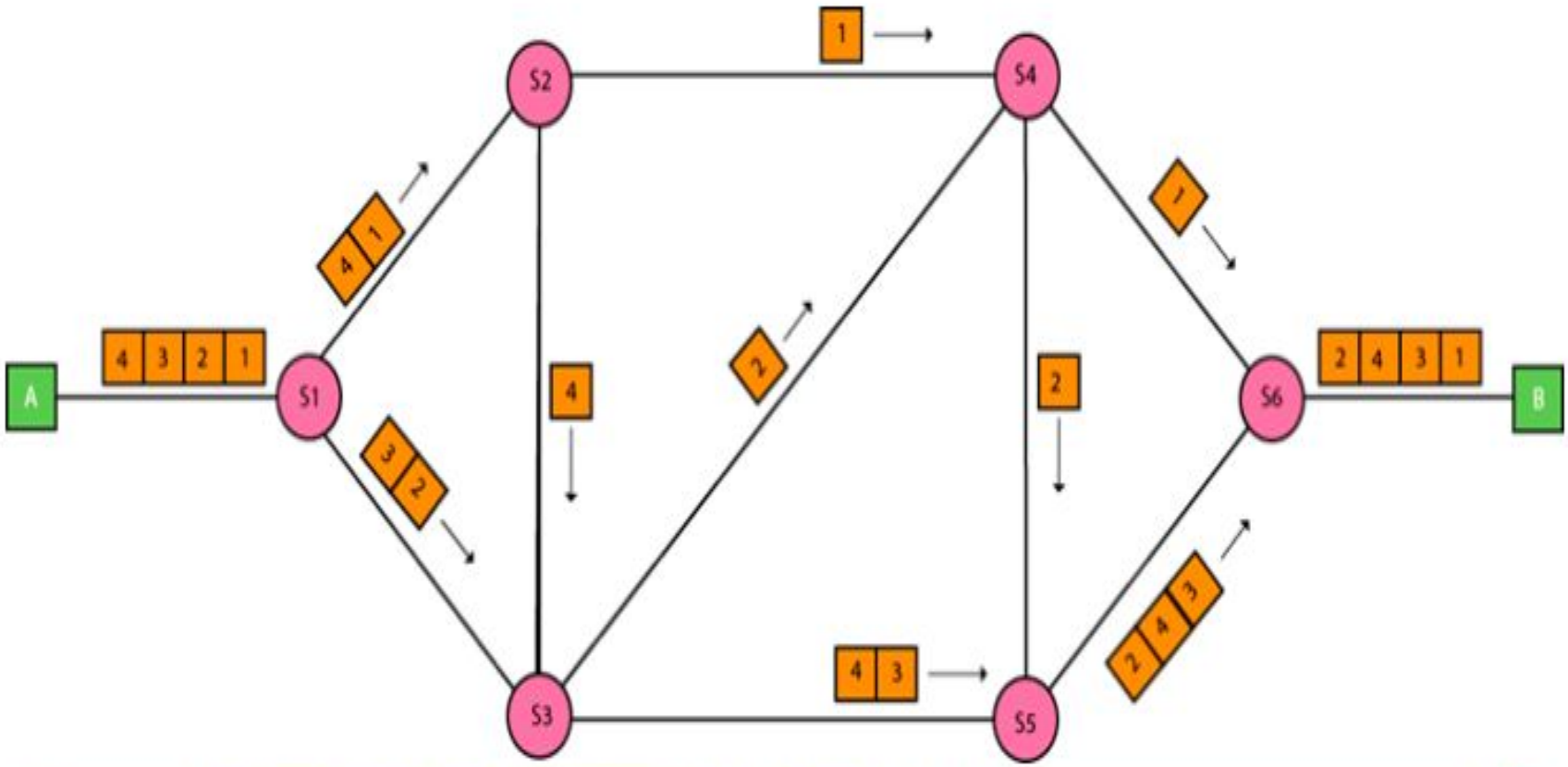
- Data channels are shared among the communicating devices that improve the efficiency of using available bandwidth.
- Traffic congestion can be reduced because the message is temporarily stored in the nodes.
- Message priority can be used to manage the network.
- The size of the message which is sent over the network can be varied. Therefore, it supports the data of unlimited size.

Disadvantages Of Message Switching

- The message switches must be equipped with sufficient storage to enable them to store the messages until the message is forwarded.
- The Long delay can occur due to the storing and forwarding facility provided by the message switching technique.

Packet Switching

- The packet switching is a switching technique in which the message is sent in one go, but it is divided into smaller pieces, and they are sent individually.
- The message splits into smaller pieces known as packets and packets are given a unique number to identify their order at the receiving end.
- Every packet contains some information in its headers such as source address, destination address and sequence number.
- Packets will travel across the network, taking the shortest path as possible.
- All the packets are reassembled at the receiving end in correct order.
- If any packet is missing or corrupted, then the message will be sent to resend the message.
- If the correct order of the packets is reached, then the acknowledgment message will be sent.



1.50

Advantages Of Packet Switching:

- **Cost-effective:** In packet switching technique, switching devices do not require massive secondary storage to store the packets, so cost is minimized to some extent. Therefore, we can say that the packet switching technique is a cost-effective technique.
- **Reliable:** If any node is busy, then the packets can be rerouted. This ensures that the Packet Switching technique provides reliable communication.
- **Efficient:** Packet Switching is an efficient technique. It does not require any established path prior to the transmission, and many users can use the same communication channel simultaneously, hence makes use of available bandwidth very efficiently.

Disadvantages Of Packet Switching:

- Packet Switching technique cannot be implemented in those applications that require low delay and high-quality services.
- The protocols used in a packet switching technique are very complex and requires high implementation cost.
- If the network is overloaded or corrupted, then it requires retransmission of lost packets. It can also lead to the loss of critical information if errors are nor recovered.

https://www.youtube.com/watch?v=oUN-s6aFMTk